

Lesson 2 Worksheet · If-Statements & Loops

Module SC101 · complete during the session, finish at the lab · Social Coding SA

Warm-up: read then predict

Before running anything, write down what each snippet prints. Then run it and check yourself.

```
for i in range(1, 6):
    print(i * 2)

total = 0
for n in [4, 7, 9]:
    total += n
print(total)
```

Exercise 1 — Pass or fail

Ask the user for a mark (0–100). Print "Pass" if it is 50 or more, otherwise "Fail". Bonus: print "Distinction" for 80+.

Exercise 2 — Counting down

Use a loop to print 10, 9, 8 ... 1, then "Lift off!". Hint: range(10, 0, -1).

Exercise 3 — Sum of evens

Add up every even number from 2 to 100 and print the answer. (You should get 2550.)

Exercise 4 — The 3-and-5 game

Print the numbers 1 to 30, but: for multiples of 3 print "Fizz", for multiples of 5 print "Buzz", and for multiples of both print "FizzBuzz". Hint: the % operator gives the remainder — 9 % 3 is 0.

Exercise 5 — Guess my number

Store a secret number. Use a while loop to keep asking the user to guess, replying "Too high" or "Too low", until they get it. Count their attempts and report it at the end.

Finished early? Change Exercise 5 so the user only gets 5 attempts.